

# **Homework #4**

## **Analysis of Algorithms**

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## Question #1

```
1 def sum_of_two_elements(S, x):
2     S = sorted(S)
3     i, j = 0, len(S) - 1
4     while i <= j:
5         if S[i] + S[j] == x:
6             return True
7         elif S[i] + S[j] < x:
8             i += 1
9         else:
10            j -= 1
11
12     return False
```

Table 1: Output for `sum_of_two_elements`

| List                                               | $x$  | $n \lg n$ solution | $n^2$ solution |
|----------------------------------------------------|------|--------------------|----------------|
| [219, 526, 87, 829, 59, 384, 849, 65, 463, 934]    | 449  | True               | True           |
| [393, 983, 584, 160, 421, 652, 41, 719, 686, 181]  | 1338 | True               | True           |
| [491, 154, 345, 508, 208, 50, 11, 183, 723, 994]   | 931  | True               | True           |
| [798, 810, 263, 786, 177, 314, 211, 708, 300, 286] | 1061 | True               | True           |
| [262, 494, 533, 38, 105, 937, 383, 625, 733, 428]  | 645  | True               | True           |
| [730, 371, 356, 591, 126, 416, 732, 84, 505, 165]  | 717  | True               | True           |
| [469, 821, 559, 492, 479, 91, 621, 55, 550, 507]   | 605  | True               | True           |
| [822, 407, 527, 858, 498, 360, 551, 532, 503, 274] | 1054 | True               | True           |
| [278, 265, 997, 628, 563, 536, 783, 817, 725, 124] | 1380 | True               | True           |
| [455, 745, 422, 274, 335, 781, 909, 867, 669, 681] | 943  | True               | True           |
| [209, 836, 717, 858, 446, 773, 507, 693, 907, 59]  | 268  | True               | True           |
| [18, 639, 711, 738, 583, 69, 714, 503, 597, 280]   | 1452 | True               | True           |
| [690, 867, 901, 558, 367, 927, 439, 590, 651, 447] | 1459 | True               | True           |
| [156, 129, 61, 363, 948, 347, 874, 914, 775, 73]   | 503  | True               | True           |
| [781, 985, 385, 523, 753, 804, 740, 7, 155, 441]   | 596  | True               | True           |
| [897, 799, 83, 402, 144, 820, 621, 22, 640, 660]   | 1420 | True               | True           |
| [478, 364, 216, 907, 638, 576, 835, 487, 571, 883] | 1051 | True               | True           |
| [193, 116, 483, 17, 363, 276, 14, 534, 145, 636]   | 162  | True               | True           |
| [626, 446, 185, 716, 514, 225, 953, 826, 758, 809] | 1072 | True               | True           |
| [268, 778, 934, 880, 347, 306, 90, 767, 626, 230]  | 1712 | True               | True           |
| [170, 591, 380, 744, 868, 242, 736, 756, 45, 798]  | 624  | False              | False          |
| [85, 354, 59, 185, 916, 42, 567, 532, 106, 285]    | 704  | False              | False          |
| [938, 218, 584, 440, 574, 748, 450, 931, 955, 588] | 909  | False              | False          |
| [930, 337, 292, 491, 668, 486, 630, 320, 91, 797]  | 17   | False              | False          |
| [277, 754, 116, 486, 75, 868, 788, 346, 326, 188]  | 847  | False              | False          |
| [951, 37, 80, 72, 515, 45, 925, 533, 626, 767]     | 346  | False              | False          |
| [529, 286, 483, 105, 283, 224, 461, 245, 447, 861] | 482  | False              | False          |
| [563, 577, 795, 911, 692, 462, 755, 311, 898, 268] | 706  | False              | False          |
| [798, 617, 27, 746, 149, 763, 546, 752, 692, 279]  | 816  | False              | False          |
| [334, 339, 184, 870, 776, 716, 375, 752, 414, 453] | 522  | False              | False          |
| [683, 969, 687, 274, 870, 566, 139, 664, 699, 325] | 17   | False              | False          |
| [641, 411, 4, 735, 742, 205, 264, 548, 331, 617]   | 604  | False              | False          |
| [876, 939, 653, 487, 148, 433, 807, 238, 848, 556] | 872  | False              | False          |
| [385, 21, 172, 598, 463, 721, 187, 670, 328, 917]  | 510  | False              | False          |
| [3, 246, 181, 435, 937, 974, 817, 109, 841, 383]   | 518  | False              | False          |
| [889, 542, 73, 455, 946, 307, 189, 988, 440, 349]  | 58   | False              | False          |
| [383, 691, 490, 938, 233, 139, 818, 231, 825, 720] | 170  | False              | False          |
| [772, 543, 127, 538, 273, 508, 924, 793, 495, 150] | 229  | False              | False          |
| [311, 843, 307, 863, 305, 602, 696, 351, 910, 776] | 42   | False              | False          |

## Question #2

1. As follows, the inversions  $(i, j)$  are:

- a) (1, 5)
- b) (2, 5)
- c) (3, 4)
- d) (3, 5)
- e) (4, 5)

$(A[i], A[j])$  are:

- a) (2, 1)
- b) (3, 5)
- c) (8, 6)
- d) (8, 1)
- e) (6, 1)

2. For a reverse sorted array, there would  $\binom{n}{2} = \frac{n(n-1)}{2}$  total inversions.

3. Divide the array recursively into half and count number of inversions in sub-arrays (an  $\lg n$  algorithm). To count all inversions, it takes  $n$  steps.  $n$  – operations  $\times$   $\lg n$  – recursive steps. Combined,  $\Theta(n \lg n)$  algorithm.

### Question #3

We recall that the geometric series, and its derivatives, state that:

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x} \quad \text{and} \quad \sum_{k=0}^{\infty} kx^k = \frac{x}{(1-x)^2} \quad (1)$$

Next, we know that our sum is as follows:

$$\sum_{k=0}^{\infty} \frac{(k-1)}{2^k}$$

Breaking these up we have the following:

$$\sum_{k=0}^{\infty} k \left(\frac{1}{2}\right)^k - \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k \quad (2)$$

Replacing Equation 2 with Equation 1, we get the following:

$$\begin{aligned} \sum_{k=0}^{\infty} k \left(\frac{1}{2}\right)^k - \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k &= \frac{1/2}{(1-1/2)^2} - \frac{1}{1-1/2} \\ &= 2 - 2 \\ &= 0 \end{aligned}$$

## Question #4

Because  $k^3$  is monotonically increasing, we have the following bounds upper bound:

$$\begin{aligned}\sum_{k=1}^n k^3 &\leq \int_1^{n+1} x^3 dx \\ &= \frac{x^4}{4} \Big|_1^{n+1} \\ &= \frac{(n+1)^4 - 1}{4}\end{aligned}$$

For the lower bound,

$$\begin{aligned}\sum_{k=1}^n k^3 &\geq \int_0^n x^3 dx \\ &= \frac{x^4}{4} \Big|_0^n \\ &= \frac{(n)^4}{4}\end{aligned}$$

So, the bounds are as follows:

$$\frac{(n)^4}{4} \leq \sum_{k=1}^n k^3 \leq \frac{(n+1)^4 - 1}{4}$$

## Question #5

We use integral approximations for all of the following.

### Problem #5.1

$$\begin{aligned} \int_0^n x^r dx &\leq \sum_{k=1}^n k^r \leq \int_1^{n+1} x^r dx \\ \frac{n^{r+1}}{r+1} &\leq \sum_{k=1}^n k^r \leq \frac{(n+1)^{r+1} - 1}{r+1} \end{aligned}$$

Therefore, the bound is  $\Theta(n^{r+1})$ .

## Problem #5.2

For the following, we take  $s$  for 1,

$$\begin{aligned}\int_0^n \lg^1 x \, dx &\leq \sum_{k=1}^n \lg^1 k \leq \int_1^{n+1} \lg^1 x \, dx \\ n(\lg(n) - 1) &\leq \sum_{k=1}^n \lg^1 k \leq (n+1)(\lg(n+1) - 1) + 1\end{aligned}$$

Therefore, the bound is  $\Theta(n \lg^1 n)$ .

For the following, we take  $s$  for 2,

$$\begin{aligned}\int_0^n \lg^2 x \, dx &\leq \sum_{k=1}^n \lg^2 k \leq \int_1^{n+1} \lg^2 x \, dx \\ n((\lg(n) - 2)\lg(n) + 2) &\leq \sum_{k=1}^n \lg^2 k \leq (n+1)((\lg(n+1) - 2)\lg(n+1) + 2) - 2\end{aligned}$$

Therefore, the bound is  $\Theta(n \lg^2 n)$ .

For the following, we take  $s$  for 3,

$$\int_0^n \lg^3 x \, dx \leq \sum_{k=1}^n \lg^3 k \leq \int_1^{n+1} \lg^3 x \, dx$$

From this, we get,

$$\begin{aligned}&n(\lg(n)((\lg(n) - 3)\lg(n) + 6) - 6) \\ &\leq \sum_{k=1}^n \lg^3 k \\ &\leq (n+1)(\lg(n+1)((\lg(n+1) - 3)\lg(n+1) + 6) - 6) + 6\end{aligned}$$

Therefore, the bound is  $\Theta(n \lg^3 n)$ .

In general, the integral requires  $s$  integration by parts, multiplying  $\lg n$   $s$  times. So, for  $\sum_{k=1}^n \lg^s k$ , the bound would be  $\Theta(n \lg^s n)$ .



### Problem #5.3

In general, we know that the lower-bound integral we have will have the form

$$\int_0^n x^r \lg^s x \, dx$$

And likewise for the upper bound. Using the same process as before (omitted due to tedious mathematics), we get the tight bound  $\Theta(n^{r+1} \lg^s n)$ .